

Spatial Exploration with a WiM for Capturing 3D Dioramic Snapshots

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ABSTRACT

Immersive Virtual Reality can be a memorable medium for story-telling and artistic expression. However, taking snapshots of VR applications is commonly only possible by taking a two-dimensional screenshot on a tethered PC. We propose to capture 3D snapshots using the interaction technique World-In-Miniature (WiM) by using virtual RGB-D cameras to create three-dimensional snapshots that can be exported and viewed outside the application. We further describe a novel interaction method to fine-tune the content shown inside the WiM with VR controllers. This project is open-source and can be accessed at <https://github.com/kyuq/3DWiMSnapshot>

1 INTRODUCTION

Immersive 3D Virtual Reality (VR) offers massive potential for story-telling and unforgettable moments in entertainment, often represented through VR video games or VR-guided stories. In conventional 2D screen-based applications, users can capture screenshots or short video sequences to keep the content in memory. Immersive three-dimensional environments are best experienced with 3D displays such as stereoscopic head-mounted displays; however, no method exists for users to persistently store a 3D snapshot.

We propose the usage of a World-in-Miniature (WiM) for taking 3D snapshots by allowing them to define a region of interest (ROI) dynamically and capturing its encapsulated 3D content into a diorama-like representation. While current methods focus on the user interaction with the WiM, we further extend on the simultaneous user interaction of the ROI and the WiM. In particular, users can grab the WiM content and fine-tune the position of the ROI from inside the WiM, or grab and fixate the WiM content and move the box of the WiM instead. To further extend the usability, we integrate dynamic size adjustment for both ROI and WiM. Similar to taking screenshots, we retrieve the rendering image of virtual RGB-D (color + depth) cameras to capture the scene dynamically without additional changes in existing shaders or render pipelines. This capturing method allows developers to use our 3D snapshots as a drop-in plugin for existing 3D applications.

Stoakley et al. [6] first conceived the concept of the World-in-Miniature for miniaturizing a virtual room in its whole and enabling users to move virtual objects by interacting with their miniature twin. Further work extended interactions of WiMs by allowing scaling and scrolling [8] for spatial navigation [2], locomotion [4], recursive interaction [1], and collaboration [3]. Trueba et al. [7] improved WiM for in-door navigation by improving clipping with a 3D in-door map. Additional work such as Voodoo-dolls [5] or Magnoramas [9] specializes in keeping the original size or magnifying the replication for added usability and supernaturally precise interactions of distant objects or regions. Most of the related work on WiMs assumes a static environment and, therefore, renders the virtual objects parameterized a second time to retrieve the miniaturized view. Only a few works [1, 9] deploy multiple cameras for duplicating the scene inside the ROI.

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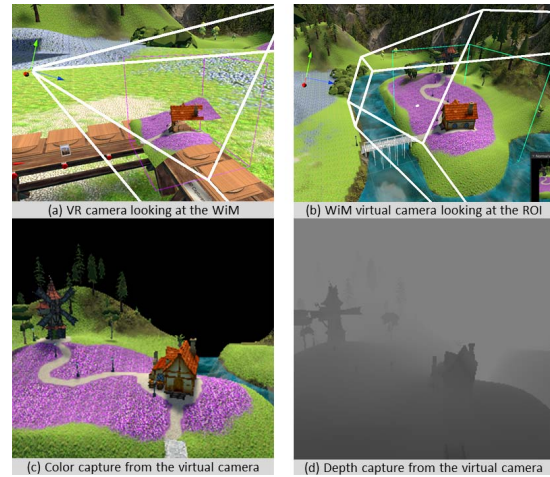


Figure 1: **Placement and View of the Virtual Camera.** We deploy virtual RGB-D cameras to capture the scene and reconstruct the 3D information to display inside the World-In-Miniature.

2 METHODS

We acquire our view onto the duplicated region by deploying additional cameras rather than duplicating the virtual objects within the ROI. The view acquisition algorithm captures 3D geometry by simulating virtual RGB-D cameras. This approach avoids requiring developers to render virtual objects a second time with scaling and clipping parameterization.

We implemented our method in Unity3D and tested it on the VR headset Vive Pro, tethered onto a laptop (Intel Core i7 @ 2.20Ghz, NVIDIA Geforce RTX 2060, 16GB RAM). We utilize in total three shaders to achieve the functionality of a WiM: (1) A post-processing shader to read the current z-buffer stored in the GPU, (2) a compute shader to convert the depth texture into an ordered, triangulated point cloud, which consists of vertices, UV coordinates and an array of triangles, and (3) a fragment shader to render the triangulated point cloud inside the scene and texturize it with the color image. We used a resolution of 1024x1024 pixels for color and depth capture. Moreover, we encoded the depth texture with 16bit per pixel, where a value of zero equals the near-clipping distance of the virtual camera and the maximum value is the far-clipping distance (see Fig. 1). The fragment shader uses Unity's compute buffers to render the virtual region without passing the point cloud geometry to the CPU and back into the GPU. During real-time interaction, a single virtual RGB-D camera captures the scene at the ROI with the transformation $T_{virtual,Cam} = T_{ROI} \bullet (T_{WiM}^{-1} \bullet T_{VR,Cam})$ where T_{WiM} and T_{ROI} are the homogeneous transformation matrix defining the pose, including rotation, translation, and scale. We create snapshots by acquiring the point cloud from the compute shader and storing the data into a Mesh container of Unity for more persistent storage and rendering structure. To improve the completeness of the snapshot, we further capture the selected ROI from the surrounding six directions. Due to the real-time capture from a simulated RGB-D camera, any an-

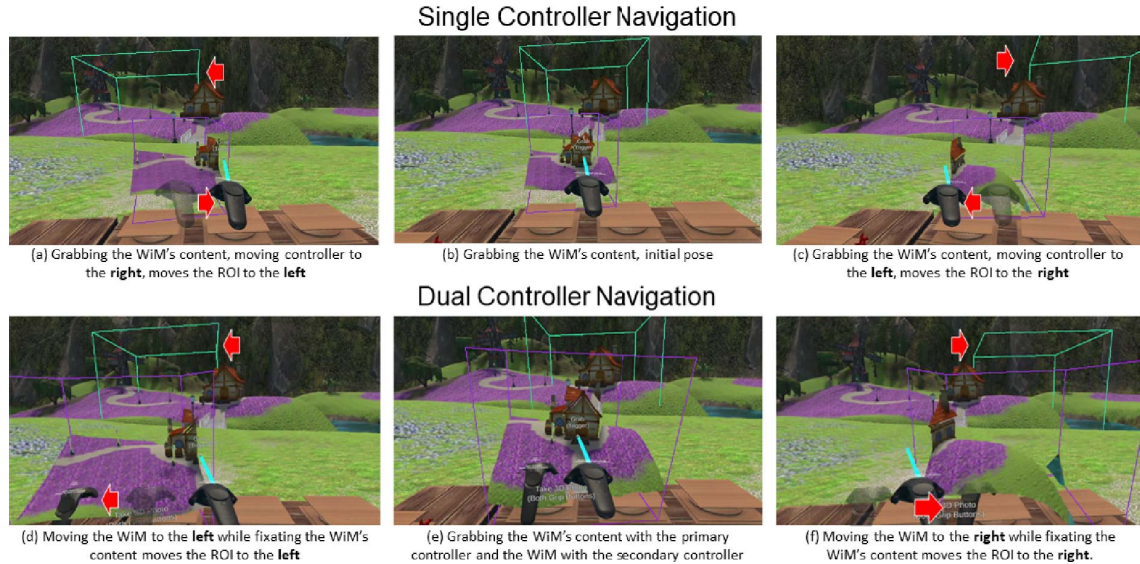


Figure 2: **Controlling the Region of Interest and the WiM.** Top row (a-c): Users operate a single controller for grabbing the content of the WiM. Controller movements in one direction move the ROI into the inverted direction. Bottom row (d-f): Users additionally use a second controller for grabbing the WiM. With both controllers, users can move the WiM instead while holding the content, in order move the ROI.

imated objects and snapshots will be visible inside the WiM. We implemented the option for users to export captured snapshots into a textured wavefront file. The system writes the triangulated point cloud from all virtual cameras into the file and additionally stores the color images for texturing the geometry.

For better navigating and adjusting the ROI for snapshots, we developed an interaction for moving the ROI from inside the WiM. In contrast to prior work with a scrolling WiM [8], we commit controller movement immediately to a change in the ROI position, as seen in Fig. 2. In particular, grabbing only the WiM content moves the ROI into the inverse direction of controller movement. The content appears to move; however, it is the ROI that is actually moving (Fig. 2 (a-c)). When users operate with two controllers, they can simultaneously move the WiM and the content of the WiM. They can now move the WiM to move the ROI along the same direction as their controller movement by fixating the content with the other controller (Fig. 2 (d-f)).

Exemplary Use-case: We integrated our method into a tranquil virtual environment where users can fulfill objectives by finding and capturing virtual items inside the world through 3D snapshots. To finish this task, they have to utilize our presented methods and our interaction technique to its fullest for exploring the environment. Users further can export their captured snapshots and open them in external 3D applications.

3 DISCUSSION & CONCLUSION

The virtual RGB-D cameras capture the scene independently of the number of virtual objects. Therefore, an advantage of this approach is its compatibility with almost any 3D environment without altering the existing rendering pipeline. Further, the added interaction technique allows for fine adjustment of the ROI while users focus on the WiM. The limitation of this approach is that geometry at the center of the ROI may be occluded by outer geometry and will be missing in the final snapshot. Further, since the virtual depth camera rely on the z-buffer generated during rendering calls, the system will not capture triangles with transparent materials and particle effects. This could be solved in future by rendering such objects with an

captureable replacement material to ensure the geometry is visible in the z-buffer.

We presented our approach of using WiMs for a comprehensive method for capturing 3D snapshots and extended on its user interaction for fine-adjusting the pose of the region of interest. As Virtual Reality expands its popularity in entertainment, we see our approach in selecting and creating snapshots as a crucial step for long lasting impressions and opens up a new type of shareable experience.

REFERENCES

- [1] A. Bluff and A. Johnston. Don't panic: Recursive interactions in a miniature metaworld. In *The 17th International Conference on Virtual-Reality Continuum and its Applications in Industry*, pp. 1–9, 2019.
- [2] D. Englmeier, W. Sajko, and A. Butz. Spherical world in miniature: Exploring the tiny planets metaphor for discrete locomotion in virtual reality. In *2021 IEEE Virtual Reality and 3D User Interfaces (VR)*, pp. 345–352. IEEE, 2021.
- [3] A. Kunert, A. Kulik, S. Beck, and B. Froehlich. Photoportals: shared references in space and time. In *17th ACM conference on Computer supported cooperative work & social computing*, pp. 1388–1399, 2014.
- [4] R. Pausch, T. Burnette, D. Brockway, and M. E. Weiblen. Navigation and locomotion in virtual worlds via flight into hand-held miniatures. In *Proceedings of the 22nd annual conference on Computer graphics and interactive techniques*, pp. 399–400, 1995.
- [5] J. S. Pierce, B. C. Stearns, and R. Pausch. Voodoo dolls: seamless interaction at multiple scales in virtual environments. In *Proceedings of the 1999 symposium on Interactive 3D graphics*, pp. 141–145, 1999.
- [6] R. Stoakley, M. J. Conway, and R. Pausch. Virtual reality on a wim: interactive worlds in miniature. In *Proceedings of the SIGCHI conference on Human factors in computing systems*, pp. 265–272, 1995.
- [7] R. Trueba, C. Andujar, and F. Argelaguet. World-in-miniature interaction for complex virtual environments. In *Innovative Design and Creation of Visual Interfaces: Advancements and Trends*, pp. 76–90, 2012.
- [8] C. A. Wingrave, Y. Hacıahmetoglu, and D. A. Bowman. Overcoming world in miniature limitations by a scaled and scrolling wim. In *3D User Interfaces (3DUI'06)*, pp. 11–16. IEEE, 2006.
- [9] K. Yu, A. Winkler, F. Pankratz, M. Lazarovici, D. Wilhelm, U. Eck, D. Roth, and N. Navab. Magnoramas: Magnifying dioramas for precise annotations in asymmetric 3d teleconsultation. In *2021 IEEE Virtual Reality and 3D User Interfaces (VR)*, pp. 392–401. IEEE, 2021.